

- 7 (a) Haemoglobin is made of two α -globin chains and two β -globin chains.

A person may have a mutation in the gene coding for β -globin. This is due to a base substitution and leads to the production of abnormal β -globin and therefore abnormal haemoglobin.

A person who is homozygous for the mutant allele will have a condition called sickle cell anaemia.

Outline the phenotypic effects of having abnormal haemoglobin in a person with sickle cell anaemia.

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- (b) **Hb^A** is the allele coding for normal β -globin.

Hb^S is the allele coding for abnormal β -globin.

The alleles are codominant.

- (i) Explain what is meant by the alleles being codominant.

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- (ii) A person who is heterozygous will have sickle cell trait but may not experience the symptoms of sickle cell anaemia.

Suggest why a person with sickle cell trait may not show the symptoms of sickle cell anaemia.

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- (iii) Construct a genetic diagram to show the possible offspring for a man and a woman who both have sickle cell trait.

parental
phenotype

sickle cell trait

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sickle cell trait

parental
genotype

gametes

offspring
genotypes

offspring
phenotypes

[3]

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